

Question Number	Answer	Additional Guidance	Mark
2(a)	Emphasis is on risk to humans • suitable risk identified (1) • suitable control of risk (1)	e.g. harmful bacteria /infection/pathogens/biohazard not contamination unqualified e.g. good aseptic technique /specific example e.g. limited lifting of lid/gloves/incubate below body temperature ignore allergies	(2)

Question Number	Answer	Additional Guidance	Mark
2(b)	Suitable working shown • correct of log 10 values (1) • correct of denominator (1) • correct answer to three sig figs (1)	ECF for MP2 and 3 4.15 & 2.95 0.602 (allow 0.301x2) 1.99 / 1.98 Correct answer (1.99/1.98) without working 3 marks Allow one mark only either for $\log_{10} 14000 - \log_{10} 900$ divided by 0.301x2 OR just 0.301x2 Allow 2 marks for a correct answer with the wrong sig figs and no working e.g. answer 1.986	(3)

Question Number	Answer	Additional Guidance	Mark
2(c)(i)	Any two from Abiotic • temperature • pH • oxygen • growth medium	Mark the first answer on each line Unless the second answer on a line is a biotic factor (negates)	(2)

Question Number	Answer	Additional Guidance	Mark
2(c)(ii)	• variable with suitable control method described ; • results are not valid / description of expected effect on the dependent variable ;	An inappropriate biotic/abiotic variable can still gain both marks E.g. incubator/AC/Thermostatically controlled waterbath Accept reduced validity	(2)

(Total for Question 2 = 9 marks)

